

Machine learning basics

Learning from examples

Machine learning is a way to build software that learns patterns from data. Instead of writing a separate rule for every situation, a developer provides examples and trains a model. The model uses the patterns it has learned to make predictions about new inputs. A common example is a system that predicts whether an email is spam based on previously labeled emails.

Supervised and unsupervised learning

In supervised learning, each training example includes a target label or value. Classification predicts a category, such as spam or not spam. Regression predicts a number, such as the expected delivery time of a package. In unsupervised learning, examples have no target labels. A clustering algorithm can group similar documents so that a person can discover common topics.

Training and evaluation

Training data helps a model learn. Validation data helps a developer choose settings, while test data provides a separate check of performance. Keeping these sets separate reduces the risk of reporting results that only reflect memorized examples. Overfitting occurs when a model performs well on its training data but poorly on new data.

From prediction to document search

Machine learning can also help people find information in documents. A search system represents text numerically and compares a question with passages from a document. For this sample project, the goal is to return relevant passages with page citations. The system does not need to generate a new answer with a language model.

Embeddings and semantic meaning

Text as a vector

An embedding is a numerical representation of an input, such as a sentence or paragraph. A text embedding model converts text into a vector: an ordered list of numbers. The number of values in the list is the embedding dimension. A document-search system uses the same compatible embedding model for both stored passages and incoming queries.

Similarity beyond exact words

Semantic search aims to find text with a similar meaning even when the wording differs. For example, the query "How do computers learn from examples?" may retrieve a passage about machine learning without requiring every query word to appear in the passage. Keyword search mainly relies on matching words, while embedding search compares numerical representations of meaning.

Comparing vectors

Cosine similarity compares the directions of two vectors. A higher cosine similarity usually indicates greater similarity within a given embedding space. Some systems use dot product or Euclidean distance instead. The comparison method should match the embedding model and index configuration. A similarity score is not a probability that a passage is correct.

Strengths and limitations

Embeddings can capture related concepts, but they do not guarantee that a result answers the question. Short, ambiguous queries can retrieve the wrong topic. Technical vocabulary, unfamiliar domains, and missing context can also reduce quality. Embeddings support retrieval; they do not verify facts or replace reading the retrieved evidence.

PDF ingestion and vector search

Extracting pages

A text-based PDF contains text that can be selected and copied. A PDF extraction library can read this text page by page. A scanned PDF usually contains images of pages and requires optical character recognition before text retrieval. This sample contains selectable text only, so it can be used to test extraction without an OCR step.

Dividing text into chunks

A chunk is a smaller passage extracted from a document. Fixed-size chunking splits text after a chosen number of characters or tokens. Paragraph-aware chunking tries to preserve complete paragraphs. Small chunks may lose useful context, while very large chunks may mix unrelated topics. Overlap repeats some text between neighboring chunks to reduce the impact of a split at a boundary.

Storing vectors and metadata

Each chunk is embedded and stored in a vector index or vector database. The record should also contain the original chunk text, document name, page number, and a stable chunk identifier. Metadata makes it possible to show a citation and inspect the source passage. If a chunk spans multiple pages, its metadata should record the page range accurately.

Searching the document

When a user submits a question, the system creates a query embedding and retrieves the nearest stored vectors. The setting K controls how many results are returned. For example, top K equal to 3 requests three candidate passages. An approximate nearest-neighbor index can speed up search in large collections, with a possible tradeoff in retrieval accuracy.

Retrieval, reranking, and evaluation

A simple user workflow

The user uploads a PDF and indexes it. The application extracts pages, creates chunks, generates embeddings, and stores the results with metadata. Once indexing is complete, the user enters a question. The interface displays ranked passages, the document name, and page citations. This workflow is an evidence search tool and does not require an LLM API or a local generation model.

Reranking candidate passages

The first retrieval step finds a set of candidate passages efficiently. A reranker then evaluates the query and each candidate together and changes their order. A cross-encoder is one possible reranking model. Reranking can improve relevance, but it adds computation and cannot recover a relevant passage that was never included in the candidate set.

Measuring Precision at K

Precision at K is the number of relevant results among the first K results divided by K. If three of the first five passages are relevant, Precision at 5 is 3 divided by 5, or 0.60. A small evaluation set should contain queries and human judgments of passage relevance. To compare chunking strategies fairly, use the same queries, embedding model, retrieval settings, and relevance rules.

Questions to try

Try searching for these questions: What is overfitting? How does semantic search differ from keyword search? Why should chunks include page metadata? When does a PDF require OCR? What does a reranker do? How is Precision at 5 calculated? Each question has supporting text in this document, making it useful for an initial manual retrieval check.